Michelle Njoroge

Computer Science Student & Developer

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Professional Profile

A detail-oriented Final Year Computer Science student at the University of Pretoria with extensive hands-on experience in full-stack web architectures, mobile UI/UX engineering, database paradigm migrations, and secure systems design. Proven track record of delivering community-focused digital solutions (MPA and Supa Neighbour) through agile team environments. Strong theoretical and practical foundation in compiler design, raw network socket programming, advanced artificial intelligence search/optimization techniques, and post-relational database management.

Technical Skills

- **Languages:** Java, Flutter/Dart, JavaScript/TypeScript, C++, Fortran 2008, Scheme/UCBLogo, Prolog, gawk
- **Frameworks & Libraries:** React Native, Node.js, Spring Boot, Firebase
- **Cloud, Databases & DevOps:** Azure, SQL, ObjectDB (OODBM), PostgreSQL (ORDBMS), BaseX (XML DB), MongoDB (NoSQL Document DB), Neo4j (Graph DB), Git version control, Doxygen automated documentation
- **Security & Networks:** TCP Raw Sockets, Snort (NIDS), Wireshark packet analysis, Cryptographic Hashing (SHA/MD5), Access Control Modeling (RBAC, Bell-LaPadula)
- **Core CS Paradigms:** Compiler Front-End/Back-End Architecture (Lexical Analysis, Parsing, Static Semantics, Code Generation), Artificial Intelligence Search & Optimization (DFS, BFS, A*, Neural Networks, Genetic Algorithms)

Technical Projects

Municipal Process Automation (MPA) | Full-Stack Developer

A web platform digitalizing municipal service delivery by connecting residents, dispatchers, and field workers in one unified system.

- Engineered a scalable web architecture using Spring Boot and Node.js for robust backend service delivery, deployed and managed on Azure.
- Implemented real-time synchronization and communication pipelines to bridge gaps between dispatchers and municipal field workers.
- **Tech Stack:** Spring Boot, Node.js, Azure, SQL, Git

Supa Neighbour Mobile Application | Frontend Developer & UI Engineer

A collaborative community-driven platform designed to streamline neighborhood assistance and task management.

- **UI/UX Engineering:** Spearheaded the cross-platform mobile frontend strategy using Flutter and React Native, developing responsive, clean, and intuitive user interfaces.
- **Full-Stack Integration:** Actively participated in backend alignment and API integration tasks using Node.js and Firebase to sync front-end interactions with real-time data streaming.
- **Documentation & Agility:** Contributed to technical system architecture layouts, component design specs, and documentation workflows using agile scrum frameworks.

Design Patterns & Documentation Framework | Core Contributor

GitHub: JoshKretschmer/COS214_Prac_GIT_Doxygen

- Developed an object-oriented software system in C++, rigorously implementing structural, behavioral, and creational design patterns.
- Automated technical documentation pipelines using Doxygen to maintain high code readability and clean architecture principles.
- Managed team workflows, branching strategies, and codebase integration through strict Git version control.

Key Coursework & Academic Practicals

- **Software Engineering:** Executed full software lifecycle engineering via extensive collaborative capstone projects, focusing on continuous integration/continuous delivery (CI/CD), microservices architecture, automated testing, and agile tracking.

- **Computer Networks:** Programmed custom client-server architectures over TCP raw sockets, explicitly engineering protocol-level handshakes and parsing network mechanics (HTTP, SMTP, POP3, FTP, LDAP).

- **Compiler Design:** Constructed a fully operational educational compiler from front-end to intermediate code generation, implementing lexical analysis, syntax parsing, symbol tables, scope-checking, type-checking, and register allocation.

- **Database Systems:** Architected and queried post-relational systems across multiple paradigms, utilizing ObjectDB (object-oriented), PostgreSQL (object-relational), BaseX (XML), MongoDB (NoSQL document), and Neo4j (graph databases).

- **Artificial Intelligence:** Implemented informed and uninformed optimization searches (DFS, BFS, A*), game-playing trees, neural networks, and genetic algorithms to solve complex problem spaces.

- **Computer Security & Ethics:** Built Role-Based Access Control (RBAC) simulators, deployed Snort NIDS rules for intrusion detection, audited memory defects, and evaluated cryptographic hashing algorithms.

- **Programming Languages:** Analyzed syntax/semantics variations and evaluated paradigms across functional recursion (Scheme/UCBLogo), logical structures (Prolog), and structural programming (Fortran 2008).

- **Trends in Multimedia:** Researched emerging digital trends, applying advanced User Experience (UX) design laws, usability testing frameworks, and multimedia production lifecycles.

Personal Projects

- **AfriCraft | Developer**
    - A digital platform designed to showcase and streamline localized artisanal commerce and marketplace interactions.
    - Designed and implemented custom data management models and full-stack interactive features to support product listings and user search workflows.
    - **GitHub Link:** michey-n/AfriCraft

- **CFC Project South | Developer**
    - A collaborative web application dedicated to presenting dynamic community-focused content and organizational initiatives.
    - Designed and implemented responsive, structured front-end web layouts to deliver an intuitive user experience across multi-page about sections.
    - **GitHub Link:** LethaboBM/CFCProjectSouth

Education

BSc in Computer Science | University of Pretoria

Expected Graduation: 2026

- **Key Coursework:** Software Engineering, Compiler Design, Database Systems, Computer Networks, Artificial Intelligence, Computer Security & Ethics, Programming Languages, Trends in Multimedia.